

An all-orders extension lemma with a prescribed two-jet, in a strict differential graded Lie algebra

Abstract

Let $\mathfrak{h}$ be a dg Lie algebra over a field of characteristic zero and let $\bar{\alpha}$ be a Maurer–Cartan element over $(s)/(s^4)$ with linear coefficient a . If the adjoint sequence $H^1(\mathfrak{h}) \rightarrow H^2(\mathfrak{h}) \rightarrow H^3(\mathfrak{h})$ of a is exact at H^2 , then $\bar{\alpha}$ extends to a Maurer–Cartan element of $s\mathfrak{h}[[s]]$ agreeing with $\bar{\alpha}$ modulo s^3 . The proof is a direct obstruction induction using only the differential, the binary bracket, and the strict Bianchi identity; every sign and index is displayed, and the initial s^4 -step is written out. This note is self-contained and makes no reference to any structure other than a strict dg Lie algebra.

1 Setting and conventions

Let k be a field of characteristic zero and $(\mathfrak{h}, \partial, [-, -])$ a dg Lie algebra over k in the cohomological convention: ∂ has degree $+1$ and satisfies $\partial^2 = 0$; the bracket has degree 0 , is graded antisymmetric, $[x, y] = -(-1)^{|x||y|}[y, x]$, satisfies the graded Jacobi identity

$$[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|}[y, [x, z]], \quad (1.1)$$

and ∂ is a degree-one derivation of it:

$$\partial[x, y] = [\partial x, y] + (-1)^{|x|}[x, \partial y]. \quad (1.2)$$

Give

$$\widehat{\mathfrak{h}} = s\mathfrak{h}[[s]] = \prod_{j \geq 1} s^j \mathfrak{h}, \quad F^r \widehat{\mathfrak{h}} = s^r \mathfrak{h}[[s]] \quad (r \geq 1),$$

the s -adic filtration; it is complete ($\widehat{\mathfrak{h}} = \varprojlim \widehat{\mathfrak{h}}/F^r$) and separated ($\bigcap_r F^r = 0$), $\partial F^r \subseteq F^r$, and $[F^r, F^{r'}] \subseteq F^{r+r'}$. For $n \geq 1$ put $\mathfrak{h}_n = \mathfrak{h} \otimes_k (s)/(s^n)$. For $\gamma \in \widehat{\mathfrak{h}}^1$ (or $\gamma \in \mathfrak{h}_n^1$) define the curvature

$$\mathcal{F}(\gamma) = \partial\gamma + \frac{1}{2}[\gamma, \gamma] \in \widehat{\mathfrak{h}}^2,$$

and $\text{MC}(-) = \{\gamma : \mathcal{F}(\gamma) = 0\}$.

2 The strict Bianchi identity

Lemma 1. *For every $\gamma \in \widehat{\mathfrak{h}}^1$,*

$$\partial\mathcal{F}(\gamma) + [\gamma, \mathcal{F}(\gamma)] = 0. \quad (2.1)$$

Proof. All operations are continuous in the filtration, so it suffices to verify the identity formally. Since $|\gamma| = 1$, graded antisymmetry gives $[\partial\gamma, \gamma] = -(-1)^{2 \cdot 1}[\gamma, \partial\gamma] = -[\gamma, \partial\gamma]$, whence, by (1.2),

$$\partial(\tfrac{1}{2}[\gamma, \gamma]) = \tfrac{1}{2}([\partial\gamma, \gamma] - [\gamma, \partial\gamma]) = -[\gamma, \partial\gamma].$$

Together with $\partial^2\gamma = 0$ this gives $\partial\mathcal{F}(\gamma) = -[\gamma, \partial\gamma]$. On the other hand, (1.1) with $x = y = z = \gamma$ reads $[\gamma, [\gamma, \gamma]] = [[\gamma, \gamma], \gamma] - [\gamma, [\gamma, \gamma]]$, while antisymmetry gives $[[\gamma, \gamma], \gamma] = -[\gamma, [\gamma, \gamma]]$; hence $3[\gamma, [\gamma, \gamma]] = 0$ and, in characteristic zero, $[\gamma, [\gamma, \gamma]] = 0$. Therefore

$$[\gamma, \mathcal{F}(\gamma)] = [\gamma, \partial\gamma] + \tfrac{1}{2}[\gamma, [\gamma, \gamma]] = [\gamma, \partial\gamma],$$

and the two displays sum to zero. $\square$

Lemma 2 (exact curvature-change formula). *For $\gamma, \delta \in \widehat{\mathfrak{h}}^1$,*

$$\mathcal{F}(\gamma + \delta) - \mathcal{F}(\gamma) = \partial\delta + [\gamma, \delta] + \tfrac{1}{2}[\delta, \delta]. \quad (2.2)$$

Proof. Expand; $[\delta, \gamma] = -(-1)^{1 \cdot 1}[\gamma, \delta] = [\gamma, \delta]$. $\square$

3 The extension lemma

Lemma 3 (fixed-two-jet extension). *Let*

$$\bar{\alpha} = sa_1 + s^2a_2 + s^3a_3 \in \text{MC}(\mathfrak{h}_4), \quad a := a_1 \in \mathfrak{h}^1.$$

Assume

$$\ker(\text{ad}_a : H^2(\mathfrak{h}) \rightarrow H^3(\mathfrak{h})) \subseteq \text{im}(\text{ad}_a : H^1(\mathfrak{h}) \rightarrow H^2(\mathfrak{h})). \quad (3.1)$$

Then there exists $\alpha_\infty \in \text{MC}(\widehat{\mathfrak{h}})$ with $\alpha_\infty \equiv \bar{\alpha} \pmod{s^3}$; in particular the coefficients of s and s^2 of α_∞ are a_1 and a_2 .

Hypothesis (3.1) is stated as an inclusion because the reverse inclusion is automatic here: by the Jacobi identity $2[a, [a, x]] = [[a, a], x]$, and the coefficient of s^2 in $\mathcal{F}(\bar{\alpha}) = 0$ gives $[a, a] = -2\partial a_2$, so for any cocycle x one has $[a, [a, x]] = -[\partial a_2, x] = -\partial[a_2, x]$, a boundary. Hence $\text{im ad}_a \subseteq \ker \text{ad}_a$ at H^2 for the linear coefficient of any Maurer–Cartan element over $(s)/(s^3)$, and (3.1) upgrades this inclusion to the equality $\ker = \text{im}$. (In the geometric application the inclusion is additionally verified by an exact matrix computation.)

Proof. Well-formedness of the linear coefficient. The coefficient of s^1 in $\mathcal{F}(\bar{\alpha}) = 0$ is $\partial a_1 = 0$, so $a = a_1 \in Z^1(\mathfrak{h})$ and ad_a is defined on cohomology; the coefficients of s^2 and s^3 give

$$\partial a_2 + \tfrac{1}{2}[a_1, a_1] = 0, \quad \partial a_3 + [a_1, a_2] = 0. \quad (3.2)$$

Initial lift. Let $\gamma^{(4)} = sa_1 + s^2a_2 + s^3a_3 \in \widehat{\mathfrak{h}}^1$ (all higher coefficients zero). Expanding and using (3.2),

$$\mathcal{F}(\gamma^{(4)}) = s^4r_4 + s^5r_5 + s^6r_6, \quad r_4 = [a_1, a_3] + \tfrac{1}{2}[a_2, a_2], \quad r_5 = [a_2, a_3], \quad r_6 = \tfrac{1}{2}[a_3, a_3]. \quad (3.3)$$

In particular $\mathcal{F}(\gamma^{(4)}) \in F^4\widehat{\mathfrak{h}}$: this is the initial case $N = 4$ of the induction below.

Induction hypothesis. For some $N \geq 4$ we have $\gamma^{(N)} \in \widehat{\mathfrak{h}}^1$ with

$$\gamma^{(N)} \equiv \bar{\alpha} \pmod{s^3}, \quad \gamma^{(N)} = sa + O(s^2), \quad \mathcal{F}(\gamma^{(N)}) \in F^N\widehat{\mathfrak{h}}. \quad (3.4)$$

Write $\mathcal{F}(\gamma^{(N)}) = s^N r_N + s^{N+1} r_{N+1} + O(s^{N+2})$ with $r_N, r_{N+1} \in \mathfrak{h}^2$.

Step 1: r_N is a cocycle. Insert the expansions $\gamma^{(N)} = \sum_{m \geq 1} s^m \gamma_m$ (with $\gamma_1 = a$) and $\mathcal{F}(\gamma^{(N)}) = \sum_{n \geq N} s^n r_n$ into the Bianchi identity (2.1) and take the coefficient of s^N : every bracket term $[\gamma_m, r_n]$ has total order $m + n \geq 1 + N > N$, so only $\partial r_N = 0$ survives.

Step 2: $[r_N] \in \ker \text{ad}_a$. The coefficient of s^{N+1} in (2.1) is

$$\partial r_{N+1} + [\gamma_1, r_N] = 0, \quad \text{i.e.} \quad [a, r_N] = -\partial r_{N+1} \in B^3(\mathfrak{h}).$$

Hence the class $[r_N] \in H^2(\mathfrak{h})$ lies in $\ker(\text{ad}_a : H^2 \rightarrow H^3)$.

Step 3: choice of the correction. By (3.1) there is a class $\beta \in H^1(\mathfrak{h})$ with $\text{ad}_a \beta = -[r_N]$. Choose a cocycle representative $v \in Z^1(\mathfrak{h})$ of β ; then $[r_N + [a, v]] = 0$ in $H^2(\mathfrak{h})$, so there exists $w \in \mathfrak{h}^1$ with

$$r_N + [a, v] = \partial w. \quad (3.5)$$

Define

$$\delta_N = s^{N-1} v - s^N w, \quad \gamma^{(N+1)} = \gamma^{(N)} + \delta_N. \quad (3.6)$$

Step 4: the correction kills r_N and disturbs nothing below. Apply (2.2) with $\delta = \delta_N$:

- $\partial \delta_N = s^{N-1} \partial v - s^N \partial w = -s^N \partial w$, since $\partial v = 0$;
- $[\gamma^{(N)}, \delta_N] = [sa, s^{N-1} v] + [\gamma^{(N)} - sa, s^{N-1} v] - [\gamma^{(N)}, s^N w] = s^N [a, v] + F^{N+1}$, since $\gamma^{(N)} - sa \in F^2$;
- $\frac{1}{2}[\delta_N, \delta_N] \in F^{2N-2} \subseteq F^{N+2}$, because $2N - 2 \geq N + 2$ for $N \geq 4$.

Hence

$$\mathcal{F}(\gamma^{(N+1)}) = \mathcal{F}(\gamma^{(N)}) + s^N ([a, v] - \partial w) + F^{N+1} = s^N (r_N + [a, v] - \partial w) + F^{N+1} = F^{N+1}$$

by (3.5). Moreover $\delta_N \in F^{N-1} \subseteq F^3$, so $\gamma^{(N+1)} \equiv \bar{\alpha} \pmod{s^3}$ and its coefficient of s is still a ; (3.4) holds with $N + 1$.

Step 5: convergence. For each fixed $j \geq 1$ the coefficient of s^j in $\gamma^{(N)}$ is constant for $N \geq j + 2$ (the step N changes orders $\geq N - 1$ only), so the sequence converges coefficientwise in the complete filtered module $\widehat{\mathfrak{h}}$ to a limit α_∞ with $\alpha_\infty \equiv \gamma^{(N)} \pmod{F^{N-1}}$ for every N . The coefficient of s^j in $\mathcal{F}(\alpha_\infty)$ is a universal finite polynomial expression in the coefficients $\alpha_1, \dots, \alpha_j$ of α_∞ ; these coincide with the corresponding coefficients of $\gamma^{(j+2)}$, whose curvature lies in F^{j+2} , so that coefficient vanishes. As j was arbitrary and $\bigcap F^r = 0$, we get $\mathcal{F}(\alpha_\infty) = 0$. The congruence $\alpha_\infty \equiv \bar{\alpha} \pmod{s^3}$ holds because every correction δ_N lies in $F^{N-1} \subseteq F^3$. $\square$

Remark 4 (the initial step, concretely). At $N = 4$ the correction is $\delta_4 = s^3 v - s^4 w$ with $r_4 = [a_1, a_3] + \frac{1}{2}[a_2, a_2]$ from (3.3). Thus the first coefficient that can change is that of s^3 , which is why the conclusion preserves $\bar{\alpha}$ modulo s^3 and not modulo s^4 . The signs in Step 4 depend only on: $\partial v = 0$; the degree-one self-pairing rule $[\delta, \gamma] = [\gamma, \delta]$ for odd elements; and the normalization $\frac{1}{2}$ in the curvature, which matches the change formula (2.2).

Remark 5 (sharpness of the preserved jet). Preservation modulo s^4 would require choosing $v = 0$ at the first step, i.e. $[r_4] = 0$ in $H^2(\mathfrak{h})$, so that a pure boundary correction $-s^4 w$ suffices. Hypothesis (3.1) yields only $[r_4] \in \ker \text{ad}_a = \text{im ad}_a$ and does not imply $[r_4] = 0$. In the geometric application, only the reduction modulo s^3 (the printed two-jet) is needed downstream, so the lemma as stated is exactly strong enough.

Remark 6 (why no higher operations are hiding here). The proof manipulates: the given filtered dg Lie structure; finitely many coefficient-extraction maps $\widehat{\mathfrak{h}} \rightarrow \mathfrak{h}$; and linear algebra in H^1, H^2, H^3 . The obstruction r_N is a coefficient of the curvature of an explicit element, not the value of a transferred n -ary operation on cohomology classes; no choice of homotopy, contraction, or reduced model on cohomology is ever made, and the element α_∞ lives in $\widehat{\mathfrak{h}}$ itself, not in its cohomology. This is the precise sense in which the lemma replaces, rather than repackages, an “all-orders vanishing of transferred curvature” argument.